A Linear-Order Resolution of the Stability Number Question for Recursive Markov Random Graphs

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Abstract

Gupte and Zhu asked whether, for fixed $p \in (0,1]$ and $r \in (0,1)$, there is a deterministic function $f(n)$ satisfying $f(n) = \Omega(n/\log n)$, $f(n) = o(n)$, and $\alpha(G^r_{n,p}) = \Theta(f(n))$ with high probability. This note gives a full solution of that question, in the negative. We prove the stronger statement

$$\alpha(G^r_{n,p}) = \Theta(n) \qquad \text{with high probability.}$$

The proof uses the terminal half of the vertex ordering. In that terminal block every possible edge has older endpoint index $j \geq n/2$, and the Markov recurrence for the edge marginals gives edge probability $O(1/n)$. Hence the terminal block has only $O(n)$ edges with high probability, and Turan's bound produces a linear independent set.

Status of the result. This is a solution, not merely partial progress, to the asymptotic-order question above under the recursive Markov random graph model of [1]. It answers the proposed sublinear-order question negatively: the stability number is already linear with high probability. The argument does not attempt to identify the sharp constant or the possible limit law for $\alpha(G^r_{n,p})/n$.

1 Model and statement of the problem

Fix constants $p \in (0,1]$ and $r \in (0,1)$, and set

$$\gamma := 1 - r \in (0,1).$$

The recursive Markov random graph $G^r_{n,p}$ has vertex set $\{v_1, \dots, v_n\}$. For $1 \leq j < i \leq n$, let X^i_j be the indicator of the edge (v_j, v_i). The edges incident to a newly added terminal vertex v_i are generated in the order $j = 1, 2, \dots, i-1$ as follows. First,

$$\mathbb{P}(X^i_1 = 1) = p.$$

For $2 \leq j < i$, the one-step transition probabilities are

$$\mathbb{P}(X^i_j = 1 \mid X^i_{j-1} = 0) = \mathbb{P}(X^i_{j-1} = 1),$$
$$\mathbb{P}(X^i_j = 1 \mid X^i_{j-1} = 1) = r\,\mathbb{P}(X^i_{j-1} = 1).$$

The construction in [1] has intra-iteration dependence but inter-iteration independence: for distinct terminal vertices i, the chains $(X^i_1, \dots, X^i_{i-1})$ are independent. This independence between terminal columns will be used only in the variance estimate in Lemma 4.

Let $\alpha(G)$ denote the stability number, i.e. the maximum cardinality of an independent set in G. Gupte and Zhu proved a lower bound of order $n/\log n$ and asked whether this is the correct order in the following sense.

Does there exist a function $f(n) = \Omega(n/\log n)$ such that $f(n) = o(n)$ and $\alpha(G^r_{n,p}) = \Theta(f(n))$ with high probability?

The theorem below shows that the answer is no.

Theorem 1 (Linear stability number)**.** *For every fixed $p \in (0,1]$ and $r \in (0,1)$, there is a constant $c = c(r) > 0$ such that*

$$\mathbb{P}\big(\alpha(G^r_{n,p}) \geq cn\big) \longrightarrow 1.$$

Consequently

$$\alpha(G^r_{n,p}) = \Theta(n) \qquad \textit{with high probability.}$$

One admissible choice is

$$c(r) = \frac{1-r}{32 + 12(1-r)}.$$

Corollary 2 (Negative answer to the sublinear-order question)**.** *There is no deterministic function $f(n)$ satisfying*

$$f(n) = \Omega\left(\frac{n}{\log n}\right), \qquad f(n) = o(n),$$

and

$$\alpha(G^r_{n,p}) = \Theta(f(n)) \qquad \textit{with high probability.}$$

If the condition $f(n) = o(n)$ is removed, then the correct asymptotic order is $f(n) = n$.

2 The marginal edge probabilities

The edge marginals are common to all terminal vertices. Define

$$q_1 := p, \qquad q_j := q_{j-1}(1 - \gamma q_{j-1}) \quad (j \geq 2).$$

Then, for every $i > j$,

$$\mathbb{P}(X^i_j = 1) = q_j.$$

Indeed, if $\mathbb{P}(X^i_{j-1} = 1) = q_{j-1}$, then the transition rule gives

$$\begin{aligned}\mathbb{P}(X^i_j = 1) &= \mathbb{P}(X^i_j = 1 \mid X^i_{j-1} = 0)\mathbb{P}(X^i_{j-1} = 0) + \mathbb{P}(X^i_j = 1 \mid X^i_{j-1} = 1)\mathbb{P}(X^i_{j-1} = 1) \\ &= q_{j-1}(1 - q_{j-1}) + rq^2_{j-1} = q_{j-1}(1 - (1-r)q_{j-1}) = q_{j-1}(1 - \gamma q_{j-1}).\end{aligned}$$

Lemma 3 (Decay of the edge marginals)**.** *For every $j \geq 1$,*

$$0 < q_j \leq \frac{1}{p^{-1} + \gamma(j-1)}.$$

In particular, if $m_n := \lceil n/2 \rceil$, then for all sufficiently large n and all $j \geq m_n$,

$$q_j \leq \frac{4}{\gamma n}.$$

Proof. The recurrence preserves positivity. Also $q_1 \le 1$, and if $0 < q_j \le 1$, then $0 < 1 - \gamma q_j \le 1$ and hence $0 < q_{j+1} \le q_j \le 1$. Thus $0 < q_j \le 1$ for all j. For $j \ge 1$,

$$\frac{1}{q_{j+1}} - \frac{1}{q_j} = \frac{1}{q_j(1-\gamma q_j)} - \frac{1}{q_j} = \frac{\gamma}{1-\gamma q_j} \ge \gamma.$$

Summing from 1 to $j-1$ gives

$$\frac{1}{q_j} \ge \frac{1}{p} + \gamma(j-1),$$

which is the first claim. If $j \ge \lceil n/2 \rceil$, then $p^{-1} + \gamma(j-1) \ge \gamma n/4$ for all sufficiently large n, and the displayed bound follows. ☐

3 A sparse terminal block

Let

$$m := m_n := \lceil n/2 \rceil, \qquad B_n := \{v_m, v_{m+1}, \dots, v_n\},$$

and let $H_n := G^r_{n,p}[B_n]$ be the subgraph induced by this terminal block. Write

$$M_n := e(H_n), \qquad N_n := |B_n| = n - m + 1.$$

Then $N_n \ge n/2$. Set

$$C_\gamma := \frac{4}{\gamma}.$$

By Lemma 3, after increasing n if necessary, every edge whose older endpoint is in B_n has marginal probability at most C_γ/n:

$$q_j \le \frac{C_\gamma}{n} \qquad (j \ge m). \tag{1}$$

For $i = m+1, \dots, n$, define

$$Y_i := \sum_{j=m}^{i-1} X^i_j.$$

Thus Y_i is the number of edges of H_n ending at v_i, and

$$M_n = \sum_{i=m+1}^{n} Y_i.$$

Lemma 4 (The terminal block has linearly many edges)**.** *With high probability,*

$$M_n \le (C_\gamma + 1)n.$$

More quantitatively, for all sufficiently large n,

$$\mathbb{P}\big(M_n > (C_\gamma + 1)n\big) \le \frac{C_\gamma + C_\gamma^2}{n}.$$

Proof. First,

$$\mathbb{E}Y_i = \sum_{j=m}^{i-1} q_j \le (i-m)\frac{C_\gamma}{n} \le C_\gamma,$$

and therefore

$$\mathbb{E}M_n \le C_\gamma n. \tag{2}$$

It remains to prove a variance bound of order n. Fix i, and let $\mathcal{F}^i_t := \sigma(X^i_1, \dots, X^i_t)$. If $m \le j < k \le i-1$, then X^i_j is $\mathcal{F}^i_{k-1}$-measurable. The Markov transition rule gives

$$\mathbb{P}(X^i_k = 1 \mid \mathcal{F}^i_{k-1}) = q_{k-1}\mathbf{1}_{\{X^i_{k-1}=0\}} + rq_{k-1}\mathbf{1}_{\{X^i_{k-1}=1\}} \le q_{k-1}.$$

Using (1),

$$\begin{aligned}\mathbb{P}(X^i_j = X^i_k = 1) &= \mathbb{E}\Big[X^i_j \mathbb{P}(X^i_k = 1 \mid \mathcal{F}^i_{k-1})\Big] \\ &\le q_j q_{k-1} \le \left(\frac{C_\gamma}{n}\right)^2. \end{aligned} \tag{3}$$

Hence

$$\begin{aligned}\mathbb{E}Y_i^2 &= \mathbb{E}Y_i + 2\sum_{m\le j<k\le i-1} \mathbb{P}(X^i_j = X^i_k = 1) \\ &\le C_\gamma + 2\binom{i-m}{2}\left(\frac{C_\gamma}{n}\right)^2 \le C_\gamma + C_\gamma^2. \end{aligned} \tag{4}$$

In particular,

$$\operatorname{Var}(Y_i) \le C_\gamma + C_\gamma^2.$$

The random variables Y_i are independent as i varies, because they are functions of distinct terminal-vertex Markov chains. Therefore

$$\operatorname{Var}(M_n) = \sum_{i=m+1}^{n} \operatorname{Var}(Y_i) \le n(C_\gamma + C_\gamma^2). \tag{5}$$

Combining (2), (5), and Chebyshev's inequality,

$$\mathbb{P}\big(M_n > (C_\gamma+1)n\big) \le \mathbb{P}\big(M_n - \mathbb{E}M_n > n\big) \le \frac{\operatorname{Var}(M_n)}{n^2} \le \frac{C_\gamma + C_\gamma^2}{n}.$$

This proves the lemma. □

4 Proof of the main theorem

We use Turan's bound in the form

$$\alpha(H) \ge \frac{|V(H)|^2}{2e(H) + |V(H)|} \tag{6}$$

for every finite graph H. On the high-probability event of Lemma 4, the graph H_n has at most $(C_\gamma+1)n$ edges and at least $n/2$ vertices. Thus (6) gives

$$\begin{aligned}\alpha(G^r_{n,p}) &\ge \alpha(H_n) \\ &\ge \frac{N_n^2}{2M_n + N_n} \\ &\ge \frac{(n/2)^2}{2(C_\gamma+1)n + n} \\ &= \frac{n}{4(2C_\gamma+3)} = \frac{\gamma}{32+12\gamma}\, n. \end{aligned}$$

Since $\gamma = 1 - r$, this proves the lower bound in Theorem 1. The upper bound $\alpha(G^r_{n,p}) \le n$ is deterministic, and therefore

$$\alpha(G^r_{n,p}) = \Theta(n) \qquad \text{with high probability.}$$

5 Excluding a sublinear order function

We now prove Corollary 2. Suppose, toward a contradiction, that there exists a deterministic $f(n) = o(n)$ such that

$$\alpha(G^r_{n,p}) = \Theta(f(n)) \qquad \text{with high probability.}$$

The upper half of this assertion gives a constant $A < \infty$ such that

$$\mathbb{P}\big(\alpha(G^r_{n,p}) \le Af(n)\big) \longrightarrow 1.$$

By Theorem 1, there is a constant $c > 0$ such that

$$\mathbb{P}\big(\alpha(G^r_{n,p}) \ge cn\big) \longrightarrow 1.$$

Therefore, for all sufficiently large n, the intersection of these two events has positive probability. For such an n there is at least one graph outcome satisfying both inequalities, so the deterministic inequality

$$cn \le Af(n)$$

holds. Hence $f(n) = \Omega(n)$, contradicting $f(n) = o(n)$. This proves the corollary.

Remark 5 (Why the global average degree is not an obstruction)**.** *Gupte and Zhu also prove that the full graph has average degree of order* $\log n$ *with high probability. The proof above is compatible with that fact: it uses only the terminal half of the ordered vertex set. Inside this terminal block, every edge has older endpoint index* $j \asymp n$*; by the recurrence for the marginals, such edge probabilities are* $O(1/n)$ *rather than* $O((\log n)/n)$ *on average over the whole graph. Thus a sparse induced subgraph on a linear number of vertices is already present.*

Remark 6 (Dependence on the model specification)**.** *The variance step uses the product construction over distinct terminal vertices. This is the "inter-iteration independence" part of the recursive model. If one were to specify only the one-step conditional probabilities within each terminal-vertex chain and not specify how different terminal-vertex chains are coupled, then those conditionals alone would not determine a unique joint law on all edges. Under the model of [1], however, the needed inter-column independence is part of the construction.*

References

[1] A. Gupte and Y. Zhu, *Large Independent Sets in Recursive Markov Random Graphs*, Mathematics of Operations Research **50**(3) (2025), 1611–1634. Published online July 12, 2024. doi:10.1287/moor.2022.0215. Preprint: arXiv:2207.04514.